Fine-tuning beyond CHSH: minimum measurement dependence for CGLMP, Mermin, and contextuality inequalities

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Relaxing the statistical independence of the measurement settings evades Bell's theorem and the contextuality theorems alike; quantifying how much must be relaxed turns "superdeterminism" and "contextual fine-tuning" from qualitative suspicions into numbers. The companion paper to this work carried out that program for the CHSH inequality; here we generalize it, with the same worst-case total-variation metric $F = \max_s \mathrm{TV}(p(\cdot \mid s) \,\|\, \rho)$ from a fixed baseline source ρ, to the CGLMP inequality for $d = 3$, the Mermin (GHZ-parity) inequalities for $n = 3$ and $n = 5$, and the Peres–Mermin square. For CGLMP $d = 3$ the CHSH headline form survives verbatim: the minimum fine-tuning to reach the achieved quantum value $T = (12 + 8\sqrt{3})/9 \approx 2.87293$ is $F^* = (T - 2)/(2\min(N, 4))$ for every N, i.e. $\sqrt{3}/9 - 1/12 \approx 0.109117$ for $N \geq 4$. Where the quantum value equals the algebraic maximum — Mermin $n = 3$ and the Peres–Mermin square — the price is the cap $\lceil N/4\rceil/N$ (saturating at 1/4) and $\lceil N/6\rceil/N$ (saturating at 1/6); the latter is, to the coverage of our search, the first exact minimum-measurement-dependence value for any contextuality scenario. For Mermin $n = 5$ frustration — every hidden state must miss at least six of the sixteen conditions — changes the answer: the exact values at $N = 2, \ldots, 6$ are $\frac{1}{2}, \frac{2}{3}, \frac{1}{2}, \frac{3}{5}, \frac{1}{2}$, with the $N = 5$ value strictly above the general lower bound $\lceil 3N/8\rceil/N$, and $\lim_{N\to\infty} F^* = \frac{3}{8} = m_{\min}/K$. We state a $K/m_{\min}$ unification principle as a conjecture whose instances are proved, and compare with Alai's recent faithful-model program for GHZ-Mermin correlations: the metrics are incommensurable, the conclusions align.

I. INTRODUCTION

The companion paper to this work [1] quantified the measurement dependence a local hidden-variable model must have to violate the CHSH inequality: it proved $S \leq 2 + 8F$ for every local CHSH model with $F = \max_{ab} \mathrm{TV}(p(\cdot \mid a, b)\|\rho)$ the worst-case total-variation (TV) distance of the conditional source from a fixed baseline ρ, computed the exact maximum $S_{\max}(N, F)$ for N-state uniform-source models, and showed that the minimum fine-tuning for the Tsirelson value is $F^*(2\sqrt{2}; N) = (\sqrt{2} - 1)/\min(N, 4)$ — flat at $(\sqrt{2} - 1)/4$ for $N \geq 4$, so that the bottleneck is the four CHSH correlators, not the size of the hidden source. The question that paper left open was whether that "four correlators" saturation is special to CHSH or is an instance of a general principle for inequalities whose algebraic maximum requires satisfying K independent conditions with a parity-type obstruction (every hidden state misses at least one).

This paper answers that question for three Bell inequalities and one contextuality scenario, in the same metric and with the same two-lemma template (a parity/obstruction lemma counting missed conditions, the row lemma of [1] bounding the per-row gain from fine-tuning, and a round-robin construction over near-miss patterns):

- **CGLMP,** $d = 3$ (Sec. III): the generalization holds *verbatim.* The achieved quantum value $T = (12 + 8\sqrt{3})/9 \approx 2.87293$ lies below the algebraic maximum 4, on the first linear segment, and $F^*(T; N) = (T - 2)/(2\min(N, 4))$ exactly, for every N.
- **Mermin (GHZ-parity),** $n = 3$ (Sec. IV): the quantum value *is* the algebraic maximum (4 on $|\mathrm{GHZ}_3\rangle$, perfect correlators), so the target sits at the cap rather than on the linear segment; the price is the cap $\lceil N/4\rceil/N$, identical to CHSH's with $K = 4$.
- **Mermin,** $n = 5$ (Sec. V): frustration — every hidden state misses at least six of the sixteen conditions ($m_{\min} = 6$) — breaks the all-but-one structure; the exact values at $N = 2, \ldots, 6$ are $\frac{1}{2}, \frac{2}{3}, \frac{1}{2}, \frac{3}{5}, \frac{1}{2}$ (the $N = 5$ value strictly above the general bound $\lceil 3N/8\rceil/N = 2/5$), with $\lim_{N\to\infty} F^* = 3/8 = m_{\min}/K$.
- **Peres–Mermin square** (Sec. VI): the state- independent contextuality analog has exactly the same shape — a parity lemma, the same row lemma, a bound $G \leq 5/6+F$, and a cap $F^*_{\mathrm{PM}}(N) = \lceil N/6\rceil/N$ saturating at 1/6 with the six lines playing the role of the four CHSH correlators.

Together these establish the generalization in one direction (CGLMP) and an identifiable failure mode in the other (GHZ-Mermin, $n \geq 5$), and they delimit the scope of the fine-tuning theory: the structural data are the triple $(K, m_{\min}, \text{balance})$, not K alone (Sec. VII). A search of the arXiv literature found no prior computation of the minimum fine-tuning in the per-setting TV metric used here for any of these scenarios (the GHZ-Mermin

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family has been treated in a different, incommensurable metric in [2], Sec. VIII); that statement, with its coverage limits, is a search result, not a proven novelty claim (Sec. IX).

Scope note. We do not treat the I3322 inequality: its quantum target 0.250875384514 . . . is a *conjectured* maximum, attained by an infinite-dimensional family of states with only a numerical upper bound on optimality, and its single-party marginal terms add biasable rows that complicate the clean K-row structure studied here. CGLMP $d \geq 4$ and Mermin $n \geq 7$ are likewise out of reach or out of scope (Sec. IX).

II. SETUP: THE METRIC, GENERALIZED

A. Models and cost

Fix a scenario with parties, settings, and outcomes as in the scenarios below, and an inequality whose value is a linear combination of the K setting blocks s that enter it. A finite model consists of:

- N hidden states $\lambda = 1, \ldots, N$ with *uniform* source $\rho(\lambda) = 1/N$;
- for each setting block s, a conditional source $p[s, \lambda] \geq 0$ with $\sum_\lambda p[s, \lambda] = 1$ (the rows $p[s, \cdot]$ are independent — no joint constraint across blocks);
- deterministic responses (per party, per local setting).

The fine-tuning cost is the exact generalization of the project's metric:

$$F = \max_s \ \mathrm{TV}\big(p(\cdot \mid s) \,\|\, \rho\big), \qquad \mathrm{TV} = \frac{1}{2} \sum_\lambda \big|p[s, \lambda] - \tfrac{1}{N}\big|. \tag{1}$$

Rows that do not enter the inequality are never biased at an optimum (biasing them costs F without helping the value), so only the K rows matter. The model-class caveats of [1] carry over: the lower-bound side does not depend on the source distribution, and stochastic responses reduce to deterministic ones by the vertex argument.

B. Generalized row lemma

Lemma 1 (row optimum for weights in $\{-1, 0, +1\}$). *Let $w_1, \ldots, w_N \in \{-1, 0, +1\}$ be the weight multiset of a row, $\mu = \frac{1}{N} \sum_\lambda w(\lambda)$ its mean, $w_{\max} = \max_\lambda w(\lambda)$, and $w_1^\uparrow \leq \cdots \leq w_N^\uparrow$ the ascending sort. The maximum of $\sum_\lambda p[\lambda]\, w(\lambda)$ over all p with $\mathrm{TV}(p \| \rho) \leq F$, at $F = k/N$, is*

$$h(k/N) = \mu + \frac{k\, w_{\max} - \sum_{i=1}^{k} w_i^\uparrow}{N}, \quad 0 \leq k \leq N-1, \qquad h(1) = w_{\max}. \tag{2}$$

For $w \in \{\pm 1\}$ this reduces to the row lemma of [1], $h = \mu + 2 \min(F, n/N)$ with n the wrong-sign count.

Proof. Write $d = p - \rho$, so d is zero-sum with $\|d\|_1 \leq 2F$. Decompose $d = d^+ - d^-$ with $d^\pm \geq 0$ and $\sum d^+ = \sum d^- = F$. Then

$$d \cdot w = \sum d^+ w - \sum d^- w \ \leq \ F\, w_{\max} - \min_{d^-} \sum d^- w.$$

The minimum of $\sum d^- w$ over $0 \leq d^- \leq 1/N$, $\sum d^- = F = k/N$ is attained by draining the smallest-w entries first (greedy; each has capacity $1/N$), giving $\sum_{i=1}^{k} w_i^\uparrow / N$. Achievability: take $d^+ = F\, e_{\arg\max}$ (capacity $(N-1)/N \geq k/N$) and d^- as in the greedy; then $p = \rho + d^+ - d^- \geq 0$ by construction. □

C. Finite-max formula and uniform bound

For fixed responses the inequality value is linear in each row $p[s, \cdot]$ and the feasible set is a product of TV balls, so the rows decouple; a hidden state's relevant data is its *pattern* — the vector of its per-row contributions. Then, exactly as in Theorem 1 of [1],

$$V_{\max}(N, F) = \max_Q \Big[V_0(Q) + \sum_r h_r(Q, F) \Big], \tag{3}$$

where Q ranges over multisets of N realizable patterns, $V_0(Q)$ is the uniform-source value, and each h_r is given by Lemma 1 (or its $\{\pm 1\}$ special case). Both are exact piecewise-linear functions of F with breakpoints at multiples of $1/N$; hence full multiset enumeration *is* the exact computation, and listing $V(k/N)$ for $k = 0, \ldots, N$ gives the complete curve. (No LP is used; the enumerations run in exact $\mathbb{Q}$ or quadratic-field arithmetic, Sec. IX.)

Proposition 1 (uniform global bound)**.** *For every multiset Q and every F,*

$$V_{\max}(N, F) \ \leq \ B + 2KF, \tag{4}$$

where $B = \max$ pattern value (the local bound).

Proof. Per row, with $d = p - \rho$ and $\|d\|_1 \leq 2F$, $|E_r - \mu_r| = |d \cdot w| \leq F\,(w_{\max} - w_{\min}) \leq 2F$; summing over the K rows and using $V_0(Q) \leq B$ (every pattern's value is at most B) gives the claim. □

Instances: CHSH (recovered from [1]), CGLMP $d = 3$ and Mermin $n = 3$ all give $2 + 8F$; Mermin $n = 5$ gives $4 + 32F$ (the $N = 5$ curve of Sec. V hits it exactly at $F = 1/5$).

Structural parameters. Two structural parameters of an inequality, plus a balance condition, govern the answers below: K = the number of extremal (row) conditions that must all be satisfied to reach the algebraic maximum A; $m_{\min}$ = the minimum number of conditions any single pattern misses (the *frustration index*); and *balance* = whether the min-miss patterns can be distributed evenly across conditions. CHSH, CGLMP $d = 3$ and Mermin $n = 3$ have $m_{\min} = 1$ ("all-but-one" near-miss structure); Mermin $n = 5$ has $m_{\min} = 6$.

III. CGLMP, $d = 3$: THE HEADLINE FORM GENERALIZES VERBATIM

A. The inequality and its constants

Two parties, two settings each (A_1, A_2 / B_1, B_2), three outcomes. The CGLMP inequality of [3] at $d = 3$ (all equalities mod 3) is

$$\begin{aligned} I_3 = &\big[P(A_1 = B_1) - P(A_1 = B_1 - 1)\big] \\ &+ \big[P(B_1 = A_2 + 1) - P(B_1 = A_2)\big] \\ &+ \big[P(A_2 = B_2) - P(A_2 = B_2 - 1)\big] \\ &+ \big[P(B_2 = A_1) - P(B_2 = A_1 - 1)\big]. \end{aligned} \tag{5}$$

- **Local bound** $I_3 \le 2$. Stated in [3] ("$I_3 \le 2$" for local realism) and independently re-proved here by exhaustive check of all 81 deterministic strategies (27 patterns), certificate `ineq_cglmp.py` Part A, cross-checked by direct brute force in `ineq_xcheck.py`.
- **Quantum value.** Collins et al. [3] give $I_3(\mathrm{QM}) = 4/(-9 + 6\sqrt{3}) = (12 + 8\sqrt{3})/9 \approx 2.87293$ for their fully-entangled state with Fourier-based measurements. The authors explicitly state that they do not have a proof that these measurements are optimal: this is a *verified lower bound on the Tsirelson bound*, and it is used here as the target T; it is *not* claimed to be the Tsirelson bound itself (Sec. IX).
- **Algebraic maximum** 4. Stated in [3]; re-proved unattainable locally here by the modular obstruction below.

B. Pattern structure: the modular obstruction ($K = 4$, $m_{\min} = 1$)

Write $d_{00} = A_1 - B_1$, $d_{01} = A_1 - B_2$, $d_{10} = A_2 - B_1$, $d_{11} = A_2 - B_2$ (mod 3). Every deterministic strategy's differences satisfy

$$d_{00} + d_{11} \equiv d_{01} + d_{10} \pmod 3, \tag{6}$$

the additive analog of CHSH's parity lemma (27 valid patterns out of 81). The four row weight vectors, in $\{-1, 0, +1\}$, are

$$w_{00}(d) = +1[d{=}0] - 1[d{=}2], \quad w_{01}(d) = +1[d{=}0] - 1[d{=}1],$$

$$w_{10}(d) = +1[d{=}2] - 1[d{=}0], \quad w_{11}(d) = +1[d{=}0] - 1[d{=}2].$$

The algebraic-maximum conditions are $d_{00} = 0$, $d_{01} = 0$, $d_{10} = 2$, $d_{11} = 0$; constraint (6) gives $0 + 0 \equiv 0 + 2$ (mod 3), false — *no pattern attains all four row conditions.* Enumeration finds exactly **four near-miss patterns, one per blocked row**, each with value $2 = B$:

$(d_{00}, d_{01}, d_{10}, d_{11})$	blocks row	value
$(2, 0, 2, 0)$	A_1, B_1	2
$(0, 1, 2, 0)$	A_1, B_2	2
$(0, 0, 0, 0)$	A_2, B_1	2
$(0, 0, 2, 2)$	A_2, B_2	2.

This is the direct CGLMP analog of CHSH's four miss-types (single cover, like CHSH); in each near-miss pattern the blocked row has weight -1 and the other three rows $+1$, so under round-robin every row contains both signs (slope 2 per row on the first segment).

C. Exact results

Theorem 1 (minimum fine-tuning at the quantum value)**.** *Let* $T = (12 + 8\sqrt{3})/9 \approx 2.87293$. *Then, for the* N*-state uniform-source class,*

$$F^*(I_3 = T;\, N) = \frac{T - 2}{2\,\min(N, 4)} \tag{7}$$

for every N*:* $F^* = (T - 2)/8 = \sqrt{3}/9 - 1/12 \approx 0.109117$ *exactly for* $N \ge 4$*;* $F^*(T; 2) = (T - 2)/4 \approx 0.21823$*;* $F^*(T; 3) = (T - 2)/6 \approx 0.14549$.

Proof. The uniform bound (4) gives $F^* \ge (T - 2)/8$ for all N. For $N \ge 4$ the round-robin construction over the four near-misses (state i carries the near-miss pattern of row i mod 4) gives $V_{\max}(N, F) \ge 2 + 8F$ on $[0, \lfloor N/4 \rfloor / N]$ (every row contains both signs, Sec. III); the target crosses there because $F^* = \sqrt{3}/9 - 1/12 < 1/7 \le \lfloor N/4 \rfloor / N$ for every $N \ge 4$ (asserted in $\mathbb{Q}(\sqrt{3})$ for $N = 4, 5, 6$; the minimum of $\lfloor N/4 \rfloor / N$ for $N \ge 4$ is $1/7$). For $N = 2, 3$ the value $(T - 2)/(2N)$ follows from the exact per-multiset segment analysis in $\mathbb{Q}(\sqrt{3})$ over *all* 378 / 3654 multisets (certificate `ineq_cglmp.py` Part B): the minimum is attained by a round-robin near-miss multiset, and no other multiset crosses T earlier. □

So the CHSH $\min(N, K)$ headline form holds *exactly*, with $K = 4$, because here the quantum-side value T is below the algebraic maximum (the target lies in the linear regime, as for CHSH). Note $F^*(\mathrm{CGLMP}, N \ge 4) \approx 0.10912 > F^*(\mathrm{CHSH}, N \ge 4) \approx 0.10355$: the CGLMP quantum point sits slightly farther along the same slope-8 segment.

Proposition 2 (cap at the algebraic maximum)**.** *The minimum fine-tuning to reach $I_3 = 4$ is $\lceil N/4 \rceil/N$ for all N.*

Proof. Write $n_r^+ = \#\{\lambda : w_r(\lambda) < +1\}$ (states blocking row r from its per-row maximum). Every pattern blocks at least one row, so $\sum_r n_r^+ \geq N$ and $\max_r n_r^+ \geq \lceil N/4 \rceil$; reaching $I_3 = 4$ forces each row supported on $\{w_r = +1\}$, hence $\mathrm{TV}_r \geq n_r^+/N$ and $F^* \geq \lceil N/4 \rceil/N$. Conversely the round-robin over the four near-misses has $\max_r n_r^+ = \lceil N/4 \rceil$ (asserted for $N = 2, \ldots, 64$; the counting argument is valid for all N), and supporting each row on its $+1$ states costs exactly n_r^+/N TV. The exhaustive minimum over all multisets equals $\lceil N/4 \rceil/N$ for $N = 2, \ldots, 6$. □

The exact curves $V_{\max}(N, k/N)$, $N = 2, \ldots, 6$ (full multiset enumeration, every breakpoint; certificate `ineq_cglmp.out`), with first segment $V_{\max}(N, F) = 2 + 2\min(N, 4)F$ on $[0, 1/N]$:

N	$V(0)$	$V(1/N)$	$V(2/N)$	$F^*(T; N)$
2	2	4	$---$	$(T-2)/4$
3	2	4	$---$	$(T-2)/6$
4	2	4	$---$	$(T-2)/8$
5	2	18/5	4	$(T-2)/8$
6	2	10/3	4	$(T-2)/8.$

IV. MERMIN (GHZ-PARITY), $n = 3$: THE CAP FORM

A. The inequality and its constants

Three parties, two settings each (X or Y), binary outcomes. The tripartite Mermin (GHZ-parity) inequality, in its standard even-weight form, is

$$M_3 = E(XXX) - E(XYY) - E(YXY) - E(YYX), \quad (8)$$

the four terms being the four tripartite perfect correlators of the $|\mathrm{GHZ}_3\rangle$ state, as catalogued for this inequality in [2] (after Hall, Phys. Rev. A 84, 022102 (2011), as cited there). Its constants are re-proved here by exact computation, not taken on authority:

- **Local bound** $B_3 = 2$**.** Proved by exhaustive deterministic-strategy check (certificate `ineq_mermin.py` Part A; max pattern value $= 2$) and independently by direct brute force over all 64 strategies with no pattern parameterization (`ineq_xcheck.py`). It equals the standard $2^{(n-1)/2}$.
- **Quantum value** 4 $=$ **algebraic maximum.** Proved by exact $\mathbb{Q}(i)$ matrix computation (`ineq_xcheck.py` Part (2); explicit Kronecker products of Pauli matrices, no trigonometric shortcut): on $|\mathrm{GHZ}_3\rangle = (|000\rangle + |111\rangle)/\sqrt{2}$ the four correlators are *perfect* — $E(XXX) = +1$ and $E(XYY) = E(YXY) = E(YYX) = -1$ — so $M_3 = 4$. Since four terms of $|\cdot| \leq 1$ cannot exceed 4, *the quantum value equals the algebraic maximum*: the key structural difference from CHSH, and the reason the headline changes form below.

B. Pattern structure: the parity obstruction ($K = 4$, $m_{\min} = 1$)

A hidden state's responses $(o_i[X], o_i[Y])$, $i = 1, \ldots, 3$, give 64 deterministic strategies. Its correlator vector over the four even-weight settings is $e_s = P \cdot \prod_{i \in S_Y} r_i$ with $(P, r_1, r_2, r_3) \in \{\pm 1\}^4$; the 16 parameter pairs carry a 2-fold redundancy ($(P, r) \sim (P, -r)$, invisible on even-weight rows), so there are exactly 8 **distinct realizable patterns** — the other 8 of the 16 possible $\{\pm 1\}^4$ vectors violate the parity product constraint

$$e_{XXX}\, e_{XYY}\, e_{YXY}\, e_{YYX} = +1$$

(each factor appears squared across the product). Each realizable pattern is realized by exactly 8 of the 64 strategies. The algebraic-maximum conditions $E(XXX) = +1$, $E(XYY) = E(YXY) = E(YYX) = -1$ have target product $(+1)(-1)^3 = -1$ while every pattern has product $+1$ — the *parity obstruction* (exact analog of the obstruction lemma of [1]). Enumeration gives: minimum miss count $= 1$; exactly **four near-miss patterns, one per condition** (32 near-miss strategies, 8 per condition); local bound $B_3 =$ max pattern value $= 2$. This is the direct Mermin analog of CHSH's four miss-types — a single cover, exactly like CHSH and CGLMP $d = 3$.

C. Exact results

Theorem 2 (minimum fine-tuning at the algebraic maximum)**.** *For the N-state uniform-source class,*

$$F^*(M_3 = 4;\, N) = \left\lceil \frac{N}{4} \right\rceil \Big/ N \quad (9)$$

for all $N \geq 2$: values $\frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \frac{2}{5}, \frac{1}{3}, \frac{2}{7}, \frac{1}{4}, \ldots,$ identical to the CHSH cap of [1] with $K = 4$.

Proof. Lower bound. Every pattern misses at least one condition, so $\sum_s n_s(Q) \geq N$ and $\max_s n_s \geq \lceil N/4 \rceil$; reaching $M_3 = 4$ forces each row fully bent (row value $+1$ requires p supported on the correct-sign states, $\mathrm{TV}_r \geq n_r/N$). *Upper bound.* Round-robin over the four near-miss patterns (state i misses condition $i \bmod 4$) gives $\max_s n_s \leq \lceil N/4 \rceil$ and $\sum_s n_s = N$ (asserted for $N = 2, \ldots, 64$; the counting argument is valid for all N), so $V_{\max}(N, \lceil N/4 \rceil/N) \geq 2 + 2(N/N) = 4$ exactly. □

The exact curves, $N = 2, \ldots, 6$ (certificate `ineq_mermin.out`; the first-segment bend values coincide with CHSH's $N = 5, 6$ bends — same K, same $m_{\min}$):

N	$V(0)$	$V(1/N)$	$V(2/N)$	$F^*(4;N)$
2	2	4	$---$	1/2
3	2	4	$---$	1/3
4	2	4	$---$	1/4
5	2	18/5	4	2/5
6	2	10/3	4	1/3.

Intermediate targets. For any $T \in (2, 4)$ lying on the linear segment (i.e. $(T-2)/8 \le \lfloor N/4 \rfloor/N$), $F^*(T;N) = (T-2)/(2\min(N,4))$ — in particular $F^*(2\sqrt{2};N) = (\sqrt{2}-1)/4$ for all $N \ge 4$, *the same number as CHSH.* So the fine-tuning price per unit of violation generalizes with $K = 4$; only the algebraic-maximum target departs from the $\min(N,K)$ form.

Why the $\min(N,K)$ *headline does not survive here.* CHSH's quantum point $2\sqrt{2} < 4$ lies on the first linear segment (slope $2K = 8$ once $N \ge K$), so $F^* = (T-B)/(2\min(N,K))$. For Mermin $n = 3$ the quantum point *is* the algebraic maximum: reaching it requires *every* row to be fully bent (not merely active), which costs the cap $\lceil N/K \rceil/N$ — a quantity that oscillates ($\frac{1}{4}, \frac{2}{5}, \frac{1}{3}, \frac{2}{7}, \dots$) rather than saturating at $N = K$. The condition structure ($K = 4$, $m_{\min} = 1$, all-but-one patterns, parity obstruction) is the same; what differs is where the target sits.

V. MERMIN, $n = 5$: FRUSTRATION GOVERNS THE PRICE

A. The inequality and its constants

Five parties, two settings each, binary outcomes: $M_5 = \sum_{|S|\text{ even}} (-1)^{|S|/2} E(S)$ over the 16 even- weight setting strings. As for $n = 3$, the constants are re-proved here by exact computation:

- **Local bound $B_5 = 4$.** Proved by exhaustive check (32 patterns; `ineq_xcheck.py`: 1024 strategies).
- **Quantum value 16 = algebraic maximum.** Proved by exact $\mathbb{Q}(i)$ computation: on $|\mathrm{GHZ}_5\rangle$ every even-weight correlator is exactly $(-1)^{|S|/2}$, so $M_5 = 16$.

B. Pattern structure: frustration ($K = 16$, $m_{\min} = 6$)

There are 32 distinct realizable patterns (2^5; same (P, r) redundancy as Sec. IV) and 16 active rows. Enumeration (certificate `ineq_mermin.py` Part D, cross-checked by `ineq_xcheck.py`) gives: **minimum miss count** $= 6$ — the all-but-one structure *fails* ($m_{\min} > 1$); exactly 16 **min-miss patterns**, with uniform incidence 6 **per condition** (each of the 16 conditions is missed by exactly 6 of the 16 min-miss patterns). The parity-type obstruction persists (no 0-miss pattern) but is *frustrated*: every state must miss at least six of the sixteen conditions.

C. First-segment slope: $2A_{\max B}$, not $2\min(K, m_{\min}N)$

On $[0, 1/N]$ the finite-max formula (3) is affine per multiset Q (value $V_0(Q) + 2F\,A(Q)$, with $A(Q) = \#\{s : 0 < n_s < N\}$ the number of active rows), so the right derivative at $F = 0$ is $2\,A_{\max B}$ where

$$\begin{aligned} A_{\max B} &= \max\{\, A(Q) : V_0(Q) = B_5 \,\} \\ &\le \min(K, m_{\min}N). \end{aligned}$$

the maximum over multisets whose patterns all achieve the local bound (each such pattern misses exactly $m_{\min} = 6$, so $\sum_s n_s = 6N$). Exact enumeration of the 16 bound-achieving patterns (`ineq_mermin.out` Part E2) gives

N	2	3	4	5	6
$A_{\max B}$	8	12	16	16	16
$\min(K, m_{\min}N)$	12	16	16	16	16.

The naive formula $2\min(K, m_{\min}N)$ is *false* at $N = 2, 3$ (it would give slopes 24, 32; the true first-segment slopes are 16, 24). At $N = 2$ the envelope bends *inside* $[0, 1/2]$: no bound-achieving Q can support the chord to $V(1/2) = 16$ (that would need $A = 12 > A_{\max B} = 8$), so a sub-bound multiset takes over before $F = 1/2$. For $N = 3, 4, 5, 6$ the bound-achieving line $B_5 + 2A_{\max B}F$ reaches $V(1/N)$ exactly (convexity of the envelope then forces it to be the whole segment): single segments of slope 24 ($N = 3$) and $32 = 2K$ ($N = 4, 5, 6$; $V(1/5) = 52/5 = 4 + 32/5$ hits the global bound $4 + 32F$ exactly). For $N \ge 4$ the miss sets of a few bound-achieving patterns already cover all 16 rows ($m_{\min}N \ge 16$), so every row is active from $F = 0$. This refines — and at small N corrects — the "$2\min(N,K)$" rule of [1], which assumed $m_{\min} = 1$.

D. Exact results

The exact curves, $N = 2, \dots, 6$ (full multiset enumeration; `ineq_mermin.out` and `ineq_mermin5_n5.out`):

N	$V(0)$	$V(1/N)$	$V(2/N)$	$V(3/N)$	$F^*(16;N)$
2	4	16	$---$	$---$	1/2
3	4	12	16	$---$	2/3
4	4	12	16	$---$	1/2
5	4	52/5	72/5	16	**3/5**
6	4	28/3	14	16	1/2.

Theorem 3 (minimum fine-tuning for Mermin $n = 5$)**.** *$F^*(M_5 = 16;N) = \min_Q \max_s n_s(Q)/N$, with exact values $F^* = \frac{1}{2}, \frac{2}{3}, \frac{1}{2}, \frac{3}{5}, \frac{1}{2}$ at $N = 2, 3, 4, 5, 6$. The $N = 5$ value $\frac{3}{5}$ lies* strictly above *the general lower bound*

$\lceil 3N/8\rceil/N = \frac{2}{5}$*; equality holds at* $N = 2, 3, 4, 6$*. For general* $N = 16m + r$*,*

$$\frac{\lceil 3N/8\rceil}{N} \;\le\; F^*(M_5 = 16; N) \;\le\; \frac{6m + r}{16m + r}, \qquad (10)$$

with equality in the lower bound at $N = 2, 3, 4, 6$ *and for all* $N \equiv 0, 1, 2 \pmod{16}$*, and*

$$\lim_{N\to\infty} F^*(M_5 = 16; N) = \tfrac{3}{8} = \frac{m_{\min}}{K}. \qquad (11)$$

Proof. Lower bound, all N. Every pattern misses at least 6 conditions, so $\sum_s n_s(Q) \ge 6N$ and $\max_s n_s \ge \lceil 6N/16\rceil = \lceil 3N/8\rceil$; reaching $M_5 = 16$ forces each row fully bent ($\mathrm{TV}_s \ge n_s/N$). *Upper bounds.* (a) For $N \equiv 0, 1, 2 \pmod{16}$: m copies of all 16 min-miss patterns (incidence $6m$ per condition) plus the remainder witnesses $r = 0, 1, 2$ (one realizable pattern; a disjoint complementary pair) give $\max_s n_s = \lceil 3N/8\rceil$ exactly (certificate Part F; the hardcoded witness masks were verified to be realizable pattern miss masks in `ineq_mermin5_n5.out` Part B, and 16 disjoint realizable pairs exist, so the $r = 2$ construction is not an isolated artifact). (b) For other residues $N = 16m+r$: trivially $F^* \le (6m+r)/(16m+r)$. The limit follows from (10); the exact $N = 5, 6$ values and full curves are the exhaustive enumeration. □

Reading. The fine-tuning price to reach the GHZ-5 algebraic maximum is governed by frustration: each state must miss at least 6 of 16 conditions, so even an infinitely large source needs $F \ge 3/8$. This is the same qualitative lesson as Alai's universal ceiling in [2] (their $F \le 1/2$ in Hall's metric), but in our TV metric with an exact asymptotic value: $\lim F^* = m_{\min}/K = 3/8$ at $n = 5$, vs the oscillating cap $\lceil N/4\rceil/N$ (liminf 1/4) at $n = 3$.

VI. THE PERES–MERMIN SQUARE: THE CONTEXTUALITY ANALOG

A. Setup: a fine-tuning problem

The Peres–Mermin (PM) square [4] is a 3×3 grid of nine binary observables on two qubits whose six lines (three rows, three columns) have operator products

$$\begin{array}{c|ccc} & C_0 & C_1 & C_2 \\ \hline R_0 & X\otimes X & X\otimes I & I\otimes X \\ R_1 & Y\otimes Y & X\otimes Z & Z\otimes X \\ R_2 & Z\otimes Z & I\otimes Z & Z\otimes I \end{array},$$

with row products $+I, +I, +I$ and column products $-I, +I, +I$, the $-I$ product being on column C_0 (any relabeling of lines is equivalent; certificate `fringe_pm.py` Part 0b verifies all six product identities and each line's commutativity by exact integer-matrix arithmetic). The product of the six line products is $-I$ while the product of the nine observables taken twice (once per row, once per column) is $+I$: a state-independent contextuality proof.

The game. A referee picks one of the six lines c uniformly; the player wins iff the product of the three preassigned outcomes on that line equals the target t_c, with $(t_0, \ldots, t_5) = (1, 1, 1, -1, 1, 1)$ (-1 on C_0, the line whose product is $-I$). A *noncontextual* (NC) model assigns to each hidden state λ a full assignment $v(\lambda) \in \{\pm 1\}^9$; by the parity lemma below no assignment satisfies all six lines, so $G \le 5/6$ at $F = 0$. A *quantum strategy* measures the three observables of the chosen line in any state — each line product is literally $\pm I$ and the target equals its sign on every line — so *quantum* $G = 1$ (certificate PM-Qa–d, exact matrices).

Measurement dependence. The source may depend on which line is measured: $p[c, \lambda] \ge 0$, row-sum 1, uniform baseline $\rho(\lambda) = 1/N$; cost $F = \max_c \mathrm{TV}(p(\cdot \mid c)\|\rho)$ — the same metric as everywhere else in this paper. The rows $p[c]$ are independent, so for a fixed state multiset Q each line can be optimized separately, exactly as in the CHSH theory. Question: what is the minimum F for an N-state NC-response model to reach $G = 1$, and what is the exact curve $G_{\max}(N, F)$? Two honest differences from CHSH, stated up front: (i) the quantum value equals the algebraic maximum (state-independent contextuality), so the interesting quantity is the *cap* rather than an intermediate quantum value — the KCBS inequality would test the intermediate case (Sec. IX); (ii) there is no even-parity restriction on states: all 512 assignments exist, of which exactly 32 odd-parity sign vectors are realizable on the six line products.

B. Parity lemma and miss-set census

Lemma 2 (PM parity)**.** *For any assignment* v*, write* $s_c(v) := t_c \prod_{cell\in c} v_{cell} \in \pm 1$ *(+1 iff line c is satisfied). Then every assignment violates an* odd *number of lines — in particular at least one.*

Proof. Each cell lies in exactly one row and one column, so

$$\prod_c s_c(v) = \frac{\prod_c \prod_{\mathrm{cell}\in c} v_{\mathrm{cell}}}{\prod_c t_c} = \frac{+1}{-1} = -1.$$

□

This is the contextuality analog of the obstruction lemma of [1]. The certificate additionally censuses all 512 assignments: exactly the 32 **odd-weight miss-sets** occur (6 single-line "near-miss" types, 20 three-line, 6 five-line), and a near-miss type exists for *every* one of the six lines (explicit examples in the log; e.g. the all-+1 assignment violates only the $-I$ line). This census is what makes the round-robin construction below possible, exactly as the four CHSH miss-types make the round-robin Q_N of [1] possible.

C. Global bound and first segment

For line c write $w_c := \#\{\lambda : s_c(\lambda) = -1\}$ and $\mu_c := (N - 2w_c)/N$. Then $P(\text{win } c) = \frac{1}{2}(1 + E_{p[c]}[s_c])$, and the row lemma of [1] (max of a linear functional over a TV ball — the only non-combinatorial ingredient of the whole project, reused verbatim) gives

$$\max_{\mathrm{TV}(p[c]\|\rho)\le F} P(\text{win } c) = 1 - \frac{w_c}{N} + \min\Big(F, \frac{w_c}{N}\Big).$$

Theorem 4 (PM global bound)**.** *For every N and every F: $G_{\max}(N, F) \le \frac{5}{6} + F$.*

Proof. For a model with miss-count vector $(w_0, \dots, w_5)$ of its state multiset Q and total $M := \sum_c w_c = \sum_\lambda \text{misscount}(\lambda)$, the row bound gives per line at most $1 - w_c/N + F$; averaging over the six lines,

$$G \;\le\; 1 - \frac{M}{6N} + F.$$

By Lemma 2, $\text{misscount}(\lambda) \ge 1$ for every state, so $M \ge N$, and therefore $G \le 5/6 + F$. $\square$

This is the exact analog of $S \le 2 + 8F$: a model-independent bound whose tightness is carried by the round-robin construction. (The CHSH version has slope $8 = 2K$ because each of $K = 4$ rows bends at rate 2 per unit TV; here each of $K = 6$ lines bends at rate 1 and the objective averages over six lines, so the total slope is 1.)

Round-robin. Let state i ($i = 0, \dots, N-1$) carry the near-miss assignment of line $i \bmod 6$. Then $w_c(Q_N) \in \{\lfloor N/6\rfloor, \lceil N/6\rceil\}$ for every c and $M = N$, so

$$G_{Q_N}(F) = \frac{5}{6} + \frac{1}{6}\sum_c \min\Big(F, \frac{w_c}{N}\Big).$$

Theorem 5 (PM first segment)**.** *For $N \ge 6$: $G_{\max}(N, F) = \frac{5}{6} + F$ on $[0, \lfloor N/6\rfloor/N]$ (all six lines active; lower bound by Q_N, upper bound by Theorem 4). For $N = 2, \dots, 5$ the round-robin has only N active lines and the whole curve is the single segment $G = \frac{5}{6} + \frac{N}{6}F$ on $[0, 1/N]$, reaching 1 at $F = 1/N$.*

Exact curves, $N = 2, \dots, 10$ (certificate PM-5). Because the objective depends on Q only through the miss-count vector w, the exact curve is the upper envelope over the finitely many *achievable* w-vectors — computed by an exact $\mathbb{Q}$ dynamic program over the 32 odd miss-set indicator vectors, then asserted to equal the round-robin curve *at every breakpoint* k/N ($k = 0, \dots, N$) for $N = 2, \dots, 10$. Grid points suffice because the envelope $E(F) = \max_w G_w(F)$ is convex (max of affines) while the round-robin value $R(F)$ is affine on each cell $[k/N, (k+1)/N]$: $h := E - R$ is convex on each cell, nonnegative everywhere (Q_N is among the maximized), and zero at both cell endpoints by the assertion; a convex function with $h \ge 0$ and $h(a) = h(b) = 0$ vanishes throughout. The breakpoint assertions therefore certify the whole curve (same technique as the exact-curve certificate of [1]). All values are exact rationals:

N	$G_{\max}(N, F)$	$F^*_{\mathrm{PM}}(N)$
2	$\frac{5}{6} + \frac{1}{3}F$ on $[0, \frac{1}{2}]$	$\frac{1}{2}$
3	$\frac{5}{6} + \frac{1}{2}F$ on $[0, \frac{1}{3}]$	$\frac{1}{3}$
4	$\frac{5}{6} + \frac{2}{3}F$ on $[0, \frac{1}{4}]$	$\frac{1}{4}$
5	$\frac{5}{6} + \frac{5}{6}F$ on $[0, \frac{1}{5}]$	$\frac{1}{5}$
6	$\frac{5}{6} + F$ on $[0, \frac{1}{6}]$	$\frac{1}{6}$
7	$\frac{5}{6} + F$; then $\frac{41}{42} + \frac{1}{6}(F - \frac{1}{7})$	$\frac{2}{7}$
8	$\frac{5}{6} + F$; then $\frac{23}{24} + \frac{1}{3}(F - \frac{1}{8})$	$\frac{1}{4}$
9	$\frac{5}{6} + F$; then $\frac{17}{18} + \frac{1}{2}(F - \frac{1}{9})$	$\frac{2}{9}$
10	$\frac{5}{6} + F$; then $\frac{14}{15} + \frac{2}{3}(F - \frac{1}{10})$	$\frac{1}{5}$.

(“then” = on the interval to F^*; every row reaches exactly 1 at F^*.) For $N \ge 6$ the first segment has slope exactly 1 — the direct analog of $S = 2 + 8F$ on $[0, 1/N]$; the bends for $N \not\equiv 0 \pmod 6$ are the analog of the CHSH bend region and are exact here for all $N \le 10$ by full enumeration.

D. The cap

Theorem 6 (PM cap)**.** *$F^*_{\mathrm{PM}}(N) := \min\{F : G_{\max}(N, F) = 1\} = \lceil N/6\rceil/N$ for every $N \ge 2$. In particular the cost* saturates at 1/6*: $F^*_{\mathrm{PM}}(6k) = 1/6$ for all k, and $F^*_{\mathrm{PM}}(N) \to 1/6$.*

Proof. Necessity. $G = 1$ forces $P(\text{win } c) = 1$ for every line c (each term is ≤ 1). By the row lemma this requires $p[c]$ supported on $\{\lambda : s_c(\lambda) = +1\}$, which — when $w_c > 0$ — requires $F \ge w_c/N$. Hence $F \ge \max_c w_c(Q)$; and by Lemma 2, $\sum_c w_c = \sum_\lambda \text{misscount}(\lambda) \ge N$, so $\max_c w_c \ge \lceil N/6\rceil$. Thus $F^*_{\mathrm{PM}}(N) \ge \lceil N/6\rceil/N$. *Sufficiency.* The round-robin Q_N has $\max_c w_c = \lceil N/6\rceil$. At $F = \lceil N/6\rceil/N$ set each row $p[c]$ uniform on its satisfying states (there are $N - w_c$ of them); then

$$\mathrm{TV}(p[c]\|\rho) = \tfrac{1}{2}\Big[\,(N - w_c)\big|\tfrac{1}{N-w_c} - \tfrac{1}{N}\big| + w_c\tfrac{1}{N}\,\Big] = \frac{w_c}{N} \le F,$$

and every line is won with probability 1, so $G = 1$. $\square$

Explicit witnesses (certificate Part PM-6, $N = 4, \dots, 8$). State i carries the near-miss assignment of line $i \bmod 6$; rows $p[c]$ uniform on satisfiers. Verified exactly in $\mathbb{Q}$: row sums 1; per-row TV $= w_c/N$; max per-row TV $= \lceil N/6\rceil/N$; $G = 1$. The $N = 4$ witness is printed in full in the log — note its elegance: the two lines that every state satisfies (C_1, C_2) need *zero* fine-tuning (uniform rows), and each of the other four rows spends exactly TV $= 1/4$.

Structural remark. The proof is a line-by-line translation of the cap theorem of [1]: parity lemma ($\sum$ misses $\ge N$); $\min_Q \max_c w_c/N$; round-robin sufficiency. The only new ingredient is the Lemma-2 census (32 odd miss-sets, all six near-miss types), which replaces CHSH's four even-parity patterns. Combined with Theorem 8 of [1] — the no-signalling CHSH cap at the algebraic maximum — the same statement now holds in two scenarios: *hiding the fine-tuning from the marginals never raises the price of reaching the algebraic maximum.*

VII. UNIFICATION: THE $(K, m_{\min}, \textbf{balance})$ PRINCIPLE

The data from the main paper [1] and the four scenarios of this paper ($F^*(A; N)$ = minimum fine-tuning to reach the algebraic maximum A; status labels in the following paragraph):

Scenario	K	$m_{\min}$	$F^*(A;N)$
CHSH	4	1	$\lceil N/4\rceil/N \to 1/4$
CGLMP $d=3$	4	1	$(T-2)/(2\min(N,4))$ at $T<A$; cap: $\lceil N/4\rceil/N \to 1/4$
Mermin $n=3$	4	1	$\lceil N/4\rceil/N \to 1/4$
PM square	6	1	$\lceil N/6\rceil/N \to 1/6$
Mermin $n=5$	16	6	$\frac{1}{2}, \frac{2}{3}, \frac{1}{2}, \frac{3}{5}, \frac{1}{2}$ $(N{=}2\ldots6)$; $\geq \lceil 3N/8\rceil/N$; $\lim F^* = 3/8$.

Conjecture 1 ($K/m_{\min}$ principle)**.** *For a parity-frustrated extremal game with K conditions, frustration index $m_{\min}$, and a near-miss state for each condition: (i) the minimum TV fine-tuning to reach the algebraic maximum A* saturates *at $m_{\min}/K$ as $N \to \infty$; (ii) when $m_{\min} = 1$ it equals $\lceil N/K\rceil/N$ exactly for all N; (iii) when the quantum value $T < A$ lies on the first linear segment, the cost to reach T is $(T-B)/(2\min(N,K))$.*

Status labels, stated explicitly: clause (ii) is **PROVED** for CHSH [1], the PM square (Theorem 6), and Mermin $n=3$ (Theorem 2); the CGLMP cap is the extrapolated instance of the proved linear-regime price. Clause (iii) is **PROVED** for CHSH and CGLMP $d=3$ (and, as an intermediate target, for Mermin $n=3$, Sec. IV). Clause (i) is **PROVED** for Mermin $n=5$ (the limit 3/8, Eq. (11)) and for all $m_{\min}=1$ cases (which saturate at $1/K$); the general case is **CONJECTURED**. The Mermin $n=5$ row delimits the pattern precisely: the *saturation value* $m_{\min}/K = 6/16 = 3/8$ survives as the limit, and the lower bound $\lceil m_{\min}N/K\rceil/N$ is tight for $N = 2,3,4,6$ and all $N \equiv 0,1,2 \pmod{16}$, but the finite-N cap form can be *strictly exceeded* — at $N=5$ the exact value $3/5$ lies above $\lceil 3\cdot 5/8\rceil/5 = 2/5$; for large m the remaining residues $r \in \{3,\ldots,15\} \pmod{16}$ are sandwiched but not pinned (Sec. IX). The structural data are therefore the triple $(K, m_{\min}, \text{balance})$, not K alone.

The $m_{\min}=1$ case reduces to exactly the two-lemma template used in both [1] and this paper: obstruction lemma ($\sum$ misses $\geq N$) + row lemma (per-condition gain bounded by the row optimum) + round-robin over near-miss types. For general $m_{\min}$ the lower bound $\lceil m_{\min}N/K\rceil/N$ is a one-line extension of the same counting argument, but the *achieving construction* needs a balanced packing of $m_{\min}$-miss states instead of round-robin — and Mermin $n=5$ shows such packings are not always available at the bound (strictly exceeded at $N=5$). That is the main open item (Sec. IX).

VIII. COMPARISON WITH ALAI'S FAITHFUL-MODEL PROGRAM

Alai [2] solve the *faithful* version of Hall's problem for GHZ-Mermin correlations (every full correlator reproduced and every proper-subset marginal vanishing), with metric the "surrendered fraction" $F := M/2$, where M is Hall's pairwise-L^1 measurement-dependence measure (no fixed baseline source). A reduction theorem shows the faithfulness constraints are free, so Hall's correlator-only threshold is promoted to the faithful value; LP optima certified by integer arithmetic give $F(n) = R/[2(R+1)]$ with $R = 2^{\lfloor (n-1)/2\rfloor}$ (the Mermin violation ratio) for $n = 3,\ldots,13$: $F(3)=1/3$, $F(5)=2/5$, $F(7)=4/9$, $F(9)=8/17$, $F(11)=16/33$, $F(13)=32/65$ (each odd-n value repeated at the following even size), with a universal ceiling $F \leq 1/2$.

Our setup is different in two ways: the metric is per-block TV from a fixed uniform source (not pairwise L^1), and the model class is plain (no faithfulness constraints). The metrics are therefore *incommensurable* — no ordering of the numbers follows. In particular the numerical coincidence Alai $F(3) = 1/3 =$ our $F^*(n=3; N=3)$ is a convention accident, not a theorem. What *does* align: both programs find that for GHZ-Mermin the minimum measurement dependence *does not vanish as the hidden source grows*, is determined by the frustration structure of the inequality rather than the source size, and admits a universal ceiling strictly below 1 (theirs $\leq 1/2$; ours $\lim 3/8$ at $n=5$, oscillating cap at $n=3$). This paper independently confirms, in a different metric, that "exact minimum measurement dependence for GHZ-Mermin" is a well-posed problem with clean structural answers — and supplies the first TV-metric values for it.

IX. SCOPE, CAVEATS, AND VERIFICATION

A. Coverage limits, stated honestly

- **CGLMP target.** $T = (12 + 8\sqrt{3})/9$ is the paper's *achieved* value, not a proven Tsirelson bound ([3] say so). If the true quantum maximum $T' > T$, then $F^*(T'; N) = (T'-2)/(2\min(N,4))$ by the same linear-segment argument whenever $(T'-2)/8 < \lfloor N/4\rfloor/N$ (for $N \geq 4$ this holds up to $T' < 2 + 8\lfloor N/4\rfloor/N \geq 3$). A prior guess "$\approx 3.34$" for the $d=3$ Tsirelson bound was *not* confirmed by any source fetched; it is recorded as a coverage limit and was not used. CGLMP $d \geq 4$ not treated.
- **Mermin $n \geq 7$.** Pattern space 2^n, rows 2^{n-1}: full enumeration intractable. The structural claims

($m_{\min}$ grows with n; GHZ attains the algebraic maximum) were re-derived here only for $n = 3, 5$ (exact $\mathbb{Q}(i)$). Alai's $F(n)$ formula (Hall metric) covers $n \leq 13$.

- **Exact curves.** Certified at $N = 2, \ldots, 6$ (all breakpoints; PM square to $N = 10$). For $N \geq 7$: first segment and cap are proved for all N (construction + global bound); the bend region is left to numerics (not computed here). The Mermin $n = 5$ residues $r \in \{3, \ldots, 15\}$ (mod 16) at large m remain sandwiched between $\lceil 3N/8 \rceil / N$ and $(6m + r)/(16m + r)$, not pinned.

- **Model-class caveats.** Constructions use the uniform source; the lower-bound side does not depend on the source distribution (TV duality, same argument as [1] — stated, not re-certified per inequality). Stochastic responses reduce to the deterministic case by the vertex argument (same as [1]) — not separately certified here. The *no-signalling sub-case* for these inequalities was *not* studied (open; for CHSH it did not change F^*, so a similar result is plausible but unproved here).

- **I3322** dropped (Sec. I); its local bound 0 was taken from the primary source's assertion, not re-proved by enumeration here.

- **KCBS analog (open).** The PM square has quantum = algebraic maximum, so it tests only the cap. The KCBS inequality [5] has a quantum value strictly between the NC bound and the algebraic maximum — the true analog of $2\sqrt{2}$ vs 4. The same two-lemma machinery applies (five contexts, binary outcomes, parity-type obstruction); the exact constants (NC bound of $\cos^2(\pi/10)$ type) must be taken from [5] before use and were *not* verified in this work.

- **Novelty search.** arXiv API term combinations ("measurement dependence" + "contextuality"; "fine-tuning" + "noncontextual", date-sorted) found no prior work computing minimum measurement dependence / fine-tuning for these scenarios in the per-setting TV-from-a-fixed-baseline metric used here; the closest hit is [2] (GHZ-Mermin Bell models, Hall's metric — a different, incommensurable metric). Two term combinations only; paywalled/non-arXiv content not covered. This is a search result with stated coverage limits, not a proven novelty claim.

B. Exact certificates (all re-runnable, all exit 0)

All results above are backed by re-runnable exact-arithmetic certificates (exact $\mathbb{Q}$ or quadratic-field arithmetic; no floating point in the proof parts; each script exits 0 iff every assertion passes):

Script	Log	Checks / runtime
`ineq_cglmp.py`	`ineq_cglmp.out`	30 / ≈ 25 s
`ineq_mermin.py`	`ineq_mermin.out`	52 / ≈ 1 s
`ineq_xcheck.py`	`ineq_xcheck2.out`	11 / < 1 s
`ineq_mermin5_n5.py`	`ineq_mermin5_n5.out`	14 / ≈ 41 s
`fringe_pm.py`	`fringe_pm.out`	88 / ≈ 1.6 s

`ineq_cglmp.py`: difference-pattern reduction; full multiset enumeration ($\mathbb{Q}$ + exact $\mathbb{Q}(\sqrt{3})$ tuples); per-multiset F^* analysis for $N = 2, 3$. `ineq_mermin.py`: (P, r) pattern parameterization; full multiset enumeration; pins the $n = 5$ curves $N = 2 \ldots 4$ and tightness of $\lceil 3N/8 \rceil / N$; enumerates $A_{\max B}$ exactly. `ineq_xcheck.py`: *independent* cross-check by direct strategy brute force (64/1024/81 strategies, no pattern parameterization) and exact $\mathbb{Q}(i)$ Kronecker computation of the GHZ correlators. `ineq_mermin5_n5.py`: independent re-implementation of the enumeration (cross-checked against the certified $N = 4$ values before use); full-curve pins $N = 5, 6$; audit of the remainder-witness realizability. `fringe_pm.py`: the PM square in full (exact $\mathbb{Q}$ / $\mathbb{Q}(\sqrt{2})$; two soft LP cross-checks, never failing the run). (The older log `ineq_xcheck.out` is stale — it records a crash of an earlier buggy draft; the current output is `ineq_xcheck2.out`.) The two route families (pattern- parameterized enumeration vs direct strategy brute force) agree on all shared pattern-structure facts, and the GHZ quantum values were computed without any shortcut formula. These certificates build on the main paper's (the row lemma, the exact-curve breakpoint technique, the round-robin witnesses), whose own certificates are listed in the supplemental material of [1].

X. CONCLUSION

The minimum-fine-tuning program of [1] generalizes exactly where the quantum point lies on the first linear segment and the obstruction has $m_{\min} = 1$ balanced near-misses (CGLMP $d = 3$), becomes the cap $\lceil N/K \rceil / N$ where the quantum point *is* the algebraic maximum (Mermin $n = 3$ with $K = 4$; the Peres–Mermin square with $K = 6$, a first exact result for a contextuality scenario), and is governed by the frustration index $m_{\min}/K$ where the obstruction is frustrated (Mermin $n = 5$, $\lim F^* = 3/8$). The $K/m_{\min}$ principle (Conjecture 1) unifies the instances, with Mermin $n = 5$ marking where the finite-N form can fail. Natural next steps: the KCBS analog (the intermediate-quantum-value case for contextuality), pinning the Mermin $n = 5$ residues $r \in \{3, \ldots, 15\}$ (mod 16), and the no-signalling sub-case for these inequalities.

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